$$\mathcal{O}(n^2)$$

More stable

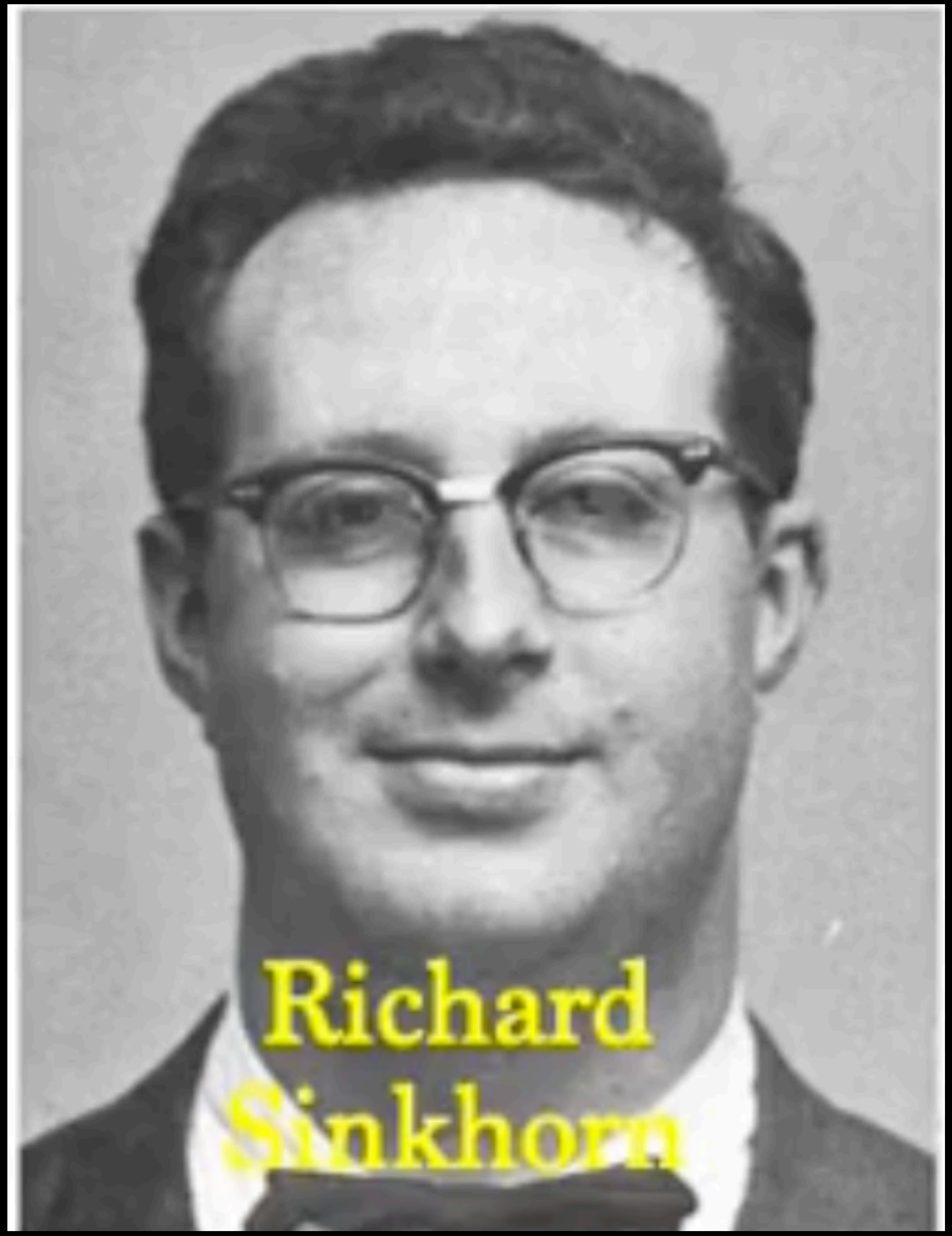
Biased

#3D251F	#3DA2B7	#ED6758	#FEC44C	#D9B37E	#AA664A
cocap	teal	salmon	citrus	beige	copper

Sinkhorn Algorithm

- Complexity:
  - Each iteration: $\mathcal{O}(n^2)$ (matrix-vector mults).
  - Typically only tens/hundreds of iterations.
- Benefits:
  - Parallelizable (GPU-friendly).
  - **More stable** numerically that classic OT, especially with log-domain tricks.
- Properties of solution:
  - **Biased**: $\mathrm{OT}_\varepsilon(\alpha, \alpha) \neq 0$ (but can be corrected)
  - Better sample complexity that unregularized OT in high dimensions

Richard D. Sinkhorn



- American mathematician(Ph.D. 1962, Wisconsin-Madison).
- Research: linear algebra, matrix theory, combinatorics, optimization.
- Sinkhorn algorithm, presented + analyzed in two papers (1964, 1967)
- Legacy: a simple, elegant iterative method that bridged pure math, numerical analysis, and today's ML applications.

A RELATIONSHIP BETWEEN ARBITRARY POSITIVE MATRICES AND DOUBLY STOCHASTIC MATRICES

BY RICHARD SINKHORN

University of Houston

1. Introduction. Suppose one observes n transitions of a Markov chain with N states and stochastic matrix $P = (p_{ij})$. The usual estimate of p_{ij} is $t_{ij} = a_{ij}/\lambda_i$ where a_{ij} is the number of transitions from i to j which are observed, and $\lambda_i = \sum_j a_{ij}$. (Cf. [1].) This amounts to a normalization of the rows of $A = (a_{ij})$, and can be expressed as a matrix equation $T = D_1 A$ where $T = (t_{ij})$ and $D_1 = \text{diag}[\lambda_1^{-1}, \cdots, \lambda_N^{-1}]$.

If it is known that the stochastic matrix P is in fact doubly stochastic, (i.e., $\sum_i p_{ij} = 1$), what then is a good estimate of T? The maximum likelihood equations are difficult to solve. One estimate which has been used (for example, by Welch [4]) is to alternately normalize the rows and columns of A, in the belief that this iterative process converges to a doubly stochastic matrix, T, which might be, in some sense, a good estimate.

CONCERNING NONNEGATIVE MATRICES AND DOUBLY STOCHASTIC MATRICES

RICHARD SINKHORN AND PAUL KNOPP

This paper is concerned with the condition for the convergence to a doubly stochastic limit of a sequence of matrices obtained from a nonnegative matrix A by alternately scaling the rows and columns of A and with the condition for the existence of diagonal matrices D_1 and D_2 with positive main diagonals such that D_1AD_2 is doubly stochastic.

The result is the following. The sequence of matrices converges to a doubly stochastic limit if and only if the matrix A contains at least one positive diagonal. A necessary and